

Trabalho 2 - Organização de computadores

Algoritmos: FFT, A* e K-means

parâmetros:

- ROB entries {32, 64, 96[FIXO], 128};
- L1D: {8KB, 16KB, 32KB[FIXO], 64KB};
- Branch Predictor{ TournamentBP[FIXO], LocalBP, BiModeBP, LTAGE}

BASE: L1I : 32KB, L2 associatividade: 8, numIQEntries 32;

FIXA: L1I : 32KB → 16 KB ; L2 associatividade: 8 → 4; numIQEntries 32 → 64

Alteração	Efeito esperado
L1I: 32 KB → 16 KB	Pode aumentar misses de instrução se o código não couber completamente na cache. Em FFT e K-Means o impacto tende a ser pequeno, pois o código é relativamente compacto. Em A* pode haver um pequeno aumento de misses devido a mais funções e desvios.
Associatividade L2: 8 → 4 vias	Aumenta a probabilidade de misses por conflito na L2. Dados que antes coexistiam na cache podem passar a competir pelas mesmas vias. FFT e K-Means tendem a ser mais sensíveis por acessarem grandes conjuntos de dados. A* pode sofrer menos dependendo do tamanho das estruturas utilizadas.
numIQEntries: 32 → 64	Aumenta a capacidade da Instruction Queue, permitindo manter mais instruções aguardando execução. O processador consegue explorar mais paralelismo de instruções (ILP), escondendo melhor latências e aumentando o IPC. FFT tende a se beneficiar mais; K-Means moderadamente; A* menos.

1) resultados FFT_ N=2048

```
Complex *x = malloc(N * sizeof(Complex));  
//      2048 × 16 bytes   = 32 768 bytes = 33 KB  
  
p.twiddle = malloc((N / 2) * sizeof(Complex));  
//      512   ×   16 bytes   = 8 192 bytes = 8 KB
```

BASE

- system.cpu.ipc_total	0.928139
- sim_seconds	0.002177
- sim_insts	4041103
- system.cpu.numCycles	4353987
- host_seconds:	30.50
- system.cpu.dcache.overall_miss_rate::cpu.data	0.005411
- system.cpu.dcache.overall_miss_rate::total	0.005411
- system.l2cache.overall_miss_rate::total	0.357749
- system.l3cache.overall_miss_rate::total	1
- system.cpu.commit.branches	221516
- system.cpu.branchPred.condPredicted	248942
- system.cpu.branchPred.condIncorrect	11152

FIXO

- system.cpu.ipc_total	0.933646
- sim_seconds	0.002164
- sim_insts	4041103
- system.cpu.numCycles	4328301
- host_seconds:	20.43
- system.cpu.dcache.overall_miss_rate::cpu.data	0.005380
- system.cpu.dcache.overall_miss_rate::total	0.005380
- system.l2cache.overall_miss_rate::total	0.354301
- system.l3cache.overall_miss_rate::total	1
- system.cpu.commit.branches	221516
- system.cpu.branchPred.condPredicted	248872
- system.cpu.branchPred.condIncorrect	10916

ROB entries

configuração 1 (32)

- system.cpu.ipc_total	0.824080
- sim_seconds	0.002452
- sim_insts	4041103
- system.cpu.numCycles	4903776
- host_seconds:	20.60
- system.cpu.dcache.overall_miss_rate::cpu.data	0.005094
- system.cpu.dcache.overall_miss_rate::total	0.005094
- system.l2cache.overall_miss_rate::total	0.352355

- system.l3cache.overall_miss_rate::total	1
- system.cpu.commit.branches	221516
- system.cpu.branchPred.condPredicted	245769
- system.cpu.branchPred.condIncorrect	10581

configuração 2 (64)

- system.cpu.ipc_total	0.916919
- sim_seconds	0.002204
- sim_insts	4041103
- system.cpu.numCycles	4407262
- host_seconds:	21.93
- system.cpu.dcache.overall_miss_rate::cpu.data	0.005353
- system.cpu.dcache.overall_miss_rate::total	0.005353
- system.l2cache.overall_miss_rate::total	0.354870
- system.l3cache.overall_miss_rate::total	1
- system.cpu.commit.branches	221516
- system.cpu.branchPred.condPredicted	245769
- system.cpu.branchPred.condIncorrect	11435

configuração 3 (128)

- system.cpu.ipc_total	0.934595
- sim_seconds	0.002162
- sim_insts	4041103
- system.cpu.numCycles	4323908
- host_seconds:	20.54
- system.cpu.dcache.overall_miss_rate::cpu.data	0.005394
- system.cpu.dcache.overall_miss_rate::total	0.005394
- system.l2cache.overall_miss_rate::total	0.354915
- system.l3cache.overall_miss_rate::total	1
- system.cpu.commit.branches	221516
- system.cpu.branchPred.condPredicted	247632
- system.cpu.branchPred.condIncorrect	10317

L1D *principal mudança no algoritmo

configuração 1 (8kb)

- system.cpu.ipc_total	0.897607
- sim_seconds	0.002251
- sim_insts	4041103
- system.cpu.numCycles	4502086
- host_seconds:	20.63
- system.cpu.dcache.overall_miss_rate::cpu.data	0.019972
- system.cpu.dcache.overall_miss_rate::total	0.019972
- system.l2cache.overall_miss_rate::total	0.117487
- system.l3cache.overall_miss_rate::total	1
- system.cpu.commit.branches	221516
- system.cpu.branchPred.condPredicted	247994
- system.cpu.branchPred.condIncorrect	10668

configuração 2 (16kb)

- system.cpu.ipc_total	0.903629
- sim_seconds	0.002236
- sim_insts	4041103
- system.cpu.numCycles	4472083
- host_seconds:	19.63
- system.cpu.dcache.overall_miss_rate::cpu.data	0.017694
- system.cpu.dcache.overall_miss_rate::total	0.017694
- system.l2cache.overall_miss_rate::total	0.12074
- system.l3cache.overall_miss_rate::total	1
- system.cpu.commit.branches	221516
- system.cpu.branchPred.condPredicted	248891
- system.cpu.branchPred.condIncorrect	10916

configuração 3 (64kb)

- system.cpu.ipc_total	0.943633
- sim_seconds	0.002141
- sim_insts	4041103
- system.cpu.numCycles	4282495
- host_seconds:	19.67
- system.cpu.dcache.overall_miss_rate::cpu.data	0.000945
- system.cpu.dcache.overall_miss_rate::total	0.000945
- system.l2cache.overall_miss_rate::total	0.950026
- system.l3cache.overall_miss_rate::total	1
- system.cpu.commit.branches	221516
- system.cpu.branchPred.condPredicted	248796
- system.cpu.branchPred.condIncorrect	10906

Branch Predictor

configuração 1 (LocalBP)

- system.cpu.ipc_total	0.912210
- sim_seconds	0.002215
- sim_insts	4041103
- system.cpu.numCycles	4430012
- host_seconds:	21.37
- system.cpu.dcache.overall_miss_rate::cpu.data	0.005330
- system.cpu.dcache.overall_miss_rate::total	0.005330
- system.l2cache.overall_miss_rate::total	0.357560
- system.l3cache.overall_miss_rate::total	1
- system.cpu.commit.branches	221516
- system.cpu.branchPred.condPredicted	248876
- system.cpu.branchPred.condIncorrect	14258

configuração 2 (BiModeBP)

- system.cpu.ipc_total	0.922936
- sim_seconds	0.002189
- sim_insts	4041103
- system.cpu.numCycles	4378531

- host_seconds:	21.12
- system.cpu.dcache.overall_miss_rate::cpu.data	0.005343
- system.cpu.dcache.overall_miss_rate::total	0.005343
- system.l2cache.overall_miss_rate::total	0.359570
- system.l3cache.overall_miss_rate::total	1
- system.cpu.commit.branches	221516
- system.cpu.branchPred.condPredicted	254959
- system.cpu.branchPred.condIncorrect	14165

configuração 3 (LTAGE)

- system.cpu.ipc_total	0.954431
- sim_seconds	0.002117
- sim_insts	4041103
- system.cpu.numCycles	4234044
- host_seconds:	19.14
- system.cpu.dcache.overall_miss_rate::cpu.data	0.005429
- system.cpu.dcache.overall_miss_rate::total	0.005429
- system.l2cache.overall_miss_rate::total	0.358258
- system.l3cache.overall_miss_rate::total	1
- system.cpu.commit.branches	221516
- system.cpu.branchPred.condPredicted	235137
- system.cpu.branchPred.condIncorrect	5850

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2) Resultado A* ****FALTA O HOST_SECONDS

BASE

- system.cpu.ipc_total	0.843536
- sim_seconds	0.015360
- system.cpu.dcache.overall_miss_rate::total	0.003516
- system.l2cache.overall_miss_rate::total	0.130087
- system.l3cache.overall_miss_rate::total	0.994164
- system.cpu.commit.branches	5023530
- system.cpu.branchPred.condPredicted	5121142
- system.cpu.branchPred.condIncorrect	15359

FIXO

- system.cpu.ipc_total	0.846392
- sim_seconds	0.015308
- sim_insts	25913518
- system.cpu.numCycles	30616451
- host_seconds	82.53
- system.cpu.dcache.overall_miss_rate::cpu.data	0.003545
- system.cpu.dcache.overall_miss_rate::total	0.003545
- system.l2cache.overall_miss_rate::total	0.132599
- system.l3cache.overall_miss_rate::total	0.973852
- system.cpu.commit.branches	5023530

- system.cpu.branchPred.condPredicted 5120338
- system.cpu.branchPred.condIncorrect 15468

CONFIGURAÇÃO 1 (ROB = 32)

- system.cpu.ipc_total 0.732781
- sim_seconds 0.017682
- sim_insts 25913518
- system.cpu.numCycles 35363274
- host_seconds 84.42
- system.cpu.dcache.overall_miss_rate::cpu.data 0.003291
- system.cpu.dcache.overall_miss_rate::total 0.003291
- system.l2cache.overall_miss_rate::total 0.138868
- system.l3cache.overall_miss_rate::total 0.975105
- system.cpu.commit.branches 5023530
- system.cpu.branchPred.condPredicted 5098438
- system.cpu.branchPred.condIncorrect 15444

CONFIGURACAO 1(ROB= 64)

- system.cpu.ipc_total 0.848630
- sim_seconds 0.015268
- sim_insts 25913518
- system.cpu.numCycles 30535706
- host_seconds 81.92
- system.cpu.dcache.overall_miss_rate::cpu.data 0.003466
- system.cpu.dcache.overall_miss_rate::total 0.003466
- system.l2cache.overall_miss_rate::total 0.132750
- system.l3cache.overall_miss_rate::total 0.973840
- system.cpu.commit.branches 5023530
- system.cpu.branchPred.condPredicted 5115164
- system.cpu.branchPred.condIncorrect 15283

CONFIGURACAO 1(ROB= 128)

- system.cpu.ipc_total 0.846538
- sim_seconds 0.015306
- sim_insts 25913518
- system.cpu.numCycles 30611171
- host_seconds 82.35
- system.cpu.dcache.overall_miss_rate::cpu.data 0.003543
- system.cpu.dcache.overall_miss_rate::total 0.003543
- system.l2cache.overall_miss_rate::total 0.132618
- system.l3cache.overall_miss_rate::total 0.974280
- system.cpu.commit.branches 5023530
- system.cpu.branchPred.condPredicted 5120255
- system.cpu.branchPred.condIncorrect 15408

CONFIGURACAO 2 (L1D = 8KB)

- system.cpu.ipc_total	0.819371
- sim_seconds	0.015813
- sim_insts	25913518
- system.cpu.numCycles	31626107
- host_seconds	87.13
- system.cpu.dcache.overall_miss_rate::cpu.data	0.015757
- system.cpu.dcache.overall_miss_rate::total	0.015757
- system.l2cache.overall_miss_rate::total	0.038058
- system.l3cache.overall_miss_rate::total	0.970646
- system.cpu.commit.branches	5023530
- system.cpu.branchPred.condPredicted	5123673
- system.cpu.branchPred.condIncorrect	15632

CONFIGURACAO 2 (L1D = 16)

- system.cpu.ipc_total	0.840267
- sim_seconds	0.015420
- sim_insts	25913518
- system.cpu.numCycles	30839607
- host_seconds	85.23
- system.cpu.dcache.overall_miss_rate::cpu.data	0.005226
- system.cpu.dcache.overall_miss_rate::total	0.005226
- system.l2cache.overall_miss_rate::total	0.088376
- system.l3cache.overall_miss_rate::total	0.971291
- system.cpu.commit.branches	5023530
- system.cpu.branchPred.condPredicted	5122773
- system.cpu.branchPred.condIncorrect	15490

CONFIGURACAO 2 (L1D = 64)

- system.cpu.ipc_total	0.852162
- sim_seconds	0.015205
- sim_insts	25913518
- system.cpu.numCycles	30408940
- host_seconds	83.05
- system.cpu.dcache.overall_miss_rate::cpu.data	0.002213
- system.cpu.dcache.overall_miss_rate::total	0.002213
- system.l2cache.overall_miss_rate::total	0.215253
- system.l3cache.overall_miss_rate::total	0.981838
- system.cpu.commit.branches	5023530
- system.cpu.branchPred.condPredicted	5119159
- system.cpu.branchPred.condIncorrect	15362

CONFIGURACAO 3 (BP = LocalBP())

- system.cpu.ipc_total	0.846710
- sim_seconds	0.015302
- sim_insts	25913518

- system.cpu.numCycles 30604953
- host_seconds 81.16
- system.cpu.dcache.overall_miss_rate::cpu.data 0.003489
- system.cpu.dcache.overall_miss_rate::total 0.003489
- system.l2cache.overall_miss_rate::total 0.135194
- system.l3cache.overall_miss_rate::total 0.974678
- system.cpu.commit.branches 5023530
- system.cpu.branchPred.condPredicted 5116331
- system.cpu.branchPred.condIncorrect 14159

CONFIGURACAO 3 (BiMode)

- system.cpu.ipc_total 0.845778
- sim_seconds 0.015319
- sim_insts 25913518
- system.cpu.numCycles 30638665
- host_seconds 80.67
- system.cpu.dcache.overall_miss_rate::cpu.data 0.003510
- system.cpu.dcache.overall_miss_rate::total 0.003510
- system.l2cache.overall_miss_rate::total 0.134571
- system.l3cache.overall_miss_rate::total 0.975333
- system.cpu.commit.branches 5023530
- system.cpu.branchPred.condPredicted 5124769
- system.cpu.branchPred.condIncorrect 18045

CONFIGURACAO 3 (LTAGE)

- system.cpu.ipc_total 0.849180
- sim_seconds 0.015258
- sim_insts 25913518
- system.cpu.numCycles 30515939
- host_seconds 84.55
- system.cpu.dcache.overall_miss_rate::cpu.data 0.003523
- system.cpu.dcache.overall_miss_rate::total 0.003523
- system.l2cache.overall_miss_rate::total 0.134224
- system.l3cache.overall_miss_rate::total 0.974896
- system.cpu.commit.branches 5023530
- system.cpu.branchPred.condPredicted 5102286
- system.cpu.branchPred.condIncorrect 13961

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3) Resultados K-Means

```
NUM_POINTS: 1000
Point points[NUM_POINTS];
// 1000 x 24 bytes = 24000 bytes = 24 KB
```


BASE

- system.cpu.ipc_total	0.854487
- sim_seconds	0.002872
- sim_insts	4907772
- system.cpu.numCycles	5743529
- host_seconds:	30.50
- system.cpu.dcache.overall_miss_rate::cpu.data	0.000390
- system.cpu.dcache.overall_miss_rate::total	0.000390
- system.l2cache.overall_miss_rate::total	0.988171
- system.l3cache.overall_miss_rate::total	1
- system.cpu.commit.branches	322914
- system.cpu.branchPred.condPredicted	356595
- system.cpu.branchPred.condIncorrect	7586

FIXO

- system.cpu.ipc_total	0.777310
- sim_seconds	0.003157
- sim_insts	4907772
- system.cpu.numCycles	6313789
- host_seconds:	32.33
- system.cpu.dcache.overall_miss_rate::cpu.data	0.000376
- system.cpu.dcache.overall_miss_rate::total	0.000376
- system.l2cache.overall_miss_rate::total	0.975741
- system.l3cache.overall_miss_rate::total	1
- system.cpu.commit.branches	322914
- system.cpu.branchPred.condPredicted	363185
- system.cpu.branchPred.condIncorrect	7997

ROB entries

configuração 1 (32)

- system.cpu.ipc_total	0.778800
- sim_seconds	0.003151
- sim_insts	4907772
- system.cpu.numCycles	6301713
- host_seconds:	30.57
- system.cpu.dcache.overall_miss_rate::cpu.data	0.000336
- system.cpu.dcache.overall_miss_rate::total	0.000336
- system.l2cache.overall_miss_rate::total	0.978261
- system.l3cache.overall_miss_rate::total	1
- system.cpu.commit.branches	322914
- system.cpu.branchPred.condPredicted	346672
- system.cpu.branchPred.condIncorrect	7927

configuração 2 (64)

- system.cpu.ipc_total	0.775867
- sim_seconds	0.003163
- sim_insts	4907772

- system.cpu.numCycles	6325531
- host_seconds:	31.71
- system.cpu.dcache.overall_miss_rate::cpu.data	0.000375
- system.cpu.dcache.overall_miss_rate::total	0.000375
- system.l2cache.overall_miss_rate::total	0.977498
- system.l3cache.overall_miss_rate::total	1
- system.cpu.commit.branches	322914
- system.cpu.branchPred.condPredicted	358289
- system.cpu.branchPred.condIncorrect	7958

configuração 3 (128)

- system.cpu.ipc_total	0.779460
- sim_seconds	0.003148
- sim_insts	4907772
- system.cpu.numCycles	6296377
- host_seconds:	32.42
- system.cpu.dcache.overall_miss_rate::cpu.data	0.000376
- system.cpu.dcache.overall_miss_rate::total	0.000376
- system.l2cache.overall_miss_rate::total	0.976682
- system.l3cache.overall_miss_rate::total	1
- system.cpu.commit.branches	322914
- system.cpu.branchPred.condPredicted	362126
- system.cpu.branchPred.condIncorrect	7655

L1D *principal mudança no algoritmo

configuração 1 (8kb)

- system.cpu.ipc_total	0.747268
- sim_seconds	0.003284
- sim_insts	4907772
- system.cpu.numCycles	6567617
- host_seconds:	32.35
- system.cpu.dcache.overall_miss_rate::cpu.data	0.010737
- system.cpu.dcache.overall_miss_rate::total	0.010737
- system.l2cache.overall_miss_rate::total	0.067044
- system.l3cache.overall_miss_rate::total	1
- system.cpu.commit.branches	322914
- system.cpu.branchPred.condPredicted	361236
- system.cpu.branchPred.condIncorrect	7500

configuração 2 (16kb)

- system.cpu.ipc_total	0.745139
- sim_seconds	0.003293
- sim_insts	4907772
- system.cpu.numCycles	6586387
- host_seconds:	34.48
- system.cpu.dcache.overall_miss_rate::cpu.data	0.010719
- system.cpu.dcache.overall_miss_rate::total	0.010719

- system.l2cache.overall_miss_rate::total	0.067012
- system.l3cache.overall_miss_rate::total	1
- system.cpu.commit.branches	322914
- system.cpu.branchPred.condPredicted	363380
- system.cpu.branchPred.condIncorrect	7994

configuração 3 (64kb)

- system.cpu.ipc_total	0.777309
- sim_seconds	0.003157
- sim_insts	4907772
- system.cpu.numCycles	6313799
- host_seconds:	32.46
- system.cpu.dcache.overall_miss_rate::cpu.data	0.000371
- system.cpu.dcache.overall_miss_rate::total	0.000371
- system.l2cache.overall_miss_rate::total	0.983696
- system.l3cache.overall_miss_rate::total	1
- system.cpu.commit.branches	322914
- system.cpu.branchPred.condPredicted	363175
- system.cpu.branchPred.condIncorrect	7992

Branch Predictor

configuração 1 (LocalBP)

- system.cpu.ipc_total	0.633856
- sim_seconds	0.003871
- sim_insts	4907772
- system.cpu.numCycles	7742724
- host_seconds:	39.19
- system.cpu.dcache.overall_miss_rate::cpu.data	0.000368
- system.cpu.dcache.overall_miss_rate::total	0.000368
- system.l2cache.overall_miss_rate::total	0.973114
- system.l3cache.overall_miss_rate::total	1
- system.cpu.commit.branches	322914
- system.cpu.branchPred.condPredicted	502982
- system.cpu.branchPred.condIncorrect	61600
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configuração 2 (BiModeBP)

- system.cpu.ipc_total	0.726786
- sim_seconds	0.003376
- sim_insts	4907772
- system.cpu.numCycles	6752707
- host_seconds:	33.33
- system.cpu.dcache.overall_miss_rate::cpu.data	0.000371
- system.cpu.dcache.overall_miss_rate::total	0.000371
- system.l2cache.overall_miss_rate::total	0.974160
- system.l3cache.overall_miss_rate::total	1
- system.cpu.commit.branches	322914
- system.cpu.branchPred.condPredicted	421544
- system.cpu.branchPred.condIncorrect	18260

configuração 3 (LTAGE)

- system.cpu.ipc_total	0.813940
- sim_seconds	0.003015
- sim_insts	4907772
- system.cpu.numCycles	6029645
- host_seconds:	30.30
- system.cpu.dcache.overall_miss_rate::cpu.data	0.000383
- system.cpu.dcache.overall_miss_rate::total	0.000383
- system.l2cache.overall_miss_rate::total	0.970537
- system.l3cache.overall_miss_rate::total	1
- system.cpu.commit.branches	322914
- system.cpu.branchPred.condPredicted	330792
- system.cpu.branchPred.condIncorrect	2123